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ORIGINAL ARTICLE

Inflation-driven instability in US sectoral betas

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Abstract
This study specifies and estimates the monthly beta coefficients as a function of annualized inflation for all sectoral exchange-traded funds (ETFs) in the USA during January 1999 to February 2024. We ensure consistency in definitions and prevent constituent overlap by using SPDR sector ETFs, which have the longest return history. The findings indicate that when inflation moves out of the 1–5% range, particularly beyond 4–5% on the upside, the functional betas across all sectors exhibit nonsensical behavior and become extremely unreliable. This behavior highlights the vulnerability of all sectoral ETFs to high inflation. Investment strategists should note that the Fed's 2% inflation target is crucial for ensuring beta's reliability as a systemic risk measure across all sectors. This finding recently highlights the need for the Fed to control inflation before cutting rates to maintain financial market stability.

Keywords Inflation · Functional coefficients · Exchange-traded funds · Sector · U.S

JEL Classification E31 · E43 · G11 · G15

Introduction

The aim of this study is to examine the impact of inflation on betas in each sector of the US equity market using a functional coefficient model of returns, providing a deeper understanding of how inflation influences systemic risk over time. Analysts have extensively debated why most central banks justifiably strive to keep inflation within narrow ranges during a business cycle (e.g., 2–3% in Australia, 2% in the USA) using explicit or implicit inflation targeting policies. This study shows how higher inflation can destabilize betas in all sectors of the US economy. In essence, the compelling focus on inflation targeting by the Fed is further validated by the findings of the present study. Although different sectors demonstrate varying levels of tolerance to inflation, a general guideline emerges: when annual inflation deviates from the 1–4% band, the beta values across all sectors become highly unstable. This inflation-induced instability severely undermines the relationship between market risk and individual exchange-traded fund's (ETF) risk, making portfolio diversification by sector very challenging. To the best of our knowledge, no previous study has systematically analyzed the time-varying impacts of inflation on sector-specific betas across different inflationary conditions, highlighting a significant gap in the literature.

Against this backdrop, we now briefly examine several studies on the effects of inflation on financial markets. High inflation disrupts resource allocation as businesses and individuals focus on hedging against inflation rather than making productive investments, affecting both economic activity and inflation uncertainty (Parikh et al. 2020). The impact of expected inflation on industry returns varies significantly. For instance, Faff and Heaney (1999) analyze long-term data to explore how equity returns in the USA, Japan, the UK, and Germany respond to inflation. They find little consistent relationship between inflation and equity returns, with expected inflation betas varying widely over time, across industries, and depending on the observation interval. Chiang (2023) examines the link between real stock returns and inflation across 12 advanced countries, focusing on overall market trends rather than sector-specific analysis. His findings highlight the negative relationship between real stock returns and increasing equity market volatility during inflationary periods, emphasizing this consistent pattern in both US and global markets. Chiang (2023) also underscores the

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substantial influence of US inflation-induced equity volatility and changes in monetary policy uncertainty on global stock markets.

In general, higher inflation often leads to higher interest rates, which can negatively impact stock valuations. This is because higher interest rates increase the discount rate used in valuing future cash flows, making current stock prices appear less attractive (Cohen et al. 2005). However, stocks can still serve as a hedge against inflation in the long run. Companies often pass on higher costs to consumers, which can help maintain their earnings and dividends, providing some protection against inflation. Modigliani and Cohn (1979) in their pioneering work argue that investors often suffer from "money illusion," where they incorrectly discount future real cash flows using nominal interest rates. This leads to falling stock prices as inflation rises because real cash flows are discounted at higher nominal rates. Cohen et al. (2005) support the money illusion hypothesis, alongside findings from other studies such as those by Fama and Schwert (1977) and Gultekin (1983), which also indicate that stock returns decline with expected and unexpected inflation changes. According to Bhamra et al. (2023), higher expected inflation disproportionately reduces real equity values for low leverage firms due to its impact on both the nominal risk-free rate and the expected growth rate of a firm's nominal cash flows (Kang and Pflueger 2015; Gomes et al. 2016). This study builds on these insights by exploring how changes in inflation influence sector-specific betas and their implied systemic risk profiles.

Econometric model

Functional coefficient regression is a semiparametric method that allows for nonlinearity and changes in the estimated coefficients over time in response to changes in other variables (Fan and Gijbels 1996; Cai et al. 2000; Huang and Shen 2004). Let us consider the relationship between the dependent variable r_t (monthly returns of a given ETF) and an explanatory variable R (monthly returns of the market). In this model, both the intercept and the beta are assumed to remain constant regardless of changes in inflation rates.

$$r_t = \alpha + \beta R_t + \varepsilon_t \tag{1}$$

where ε_t is the stochastic residual and β is the average response of a given sectoral ETF to changes in the market return. In other words, β represents the systematic risk of an ETF compared to the market, capturing the risk that cannot be diversified away. Following several studies (e.g., Aggrawal and Waggle 2010; Auer and Schuhmacher 2015), and for simplicity, we do not subtract the risk-free rate from both sides of the equation, as it has little impact on our results. One alternative is to let β be influenced inflation:

$$r_t = \alpha(Z_t) + \beta(Z_t)R_t + \varepsilon_t \tag{2}$$

where Z_t is annualized rate inflation in month t. As can be seen, the interpretation of β is no longer straightforward as its value depends on the value of Z_t. As inflation changes, so does the marginal effect of R_t on r_t. However, we can apply a local polynomial regression in a functional coefficient model using Taylor's expansion and a kernel function. This approach entails estimating a local regression with kernel-weighted squared residuals for any given rate of inflation (z). By estimating this regression across all Z_t values, we can map out the functional coefficients. By adopting a similar recursive function proposed in S&P Global (2024, User's Guide 2, p. 644), both the α and β coefficients can be expressed in a linear Taylor expansion at a given z as follows:

$$S(z) = \sum_{t=1}^{T} \{r_t - (\alpha + \beta R_t) - (\alpha^{(1)} + \beta^{(1)} R_t)(Z_t - z)\}^2 . K\left(\frac{Z_t - z}{h}\right) \tag{3}$$

Minimizing this objective function yields estimates of the coefficients at the specified point:

$$Min_{\alpha,\alpha^{(1)}\beta,\beta^{(1)}} S(z) \rightarrow \alpha(z), \beta(z), \alpha^{(1)}(z), \beta^{(1)}(z) \tag{4}$$

We can perform this minimization for different values of z. As a result, the functional coefficient is made up of a series of coefficients estimated for all values of Z_t. The objective function changes with the kernel bandwidth, ranging from zero to infinity. When $h \rightarrow 0$, the functional coefficient estimator closely fits the data, resulting in small bias but large variance, making it sensitive to noise. Conversely, when $h \rightarrow \infty$, the estimator becomes the mean of r_t, leading to large bias but small variance, as it becomes less responsive to individual data points and more stable. The optimal balance lies between these two extremes. We utilize the Epanechnikov function which is an optimal kernel in nonparametric statistics for smoothing and density estimation as follows:

$$K(u) = \frac{3}{4}(1 - u^2) I(\lfloor u \rfloor \leq 1) \tag{5}$$

where $u_t = \frac{Z_t - z}{h}$ and $K(u_t) = 0$ when $|u_t| > 1$. By selecting an optimal bandwidth this way, we can strike a balance between bias (how far our estimate is from the true value) and variance (how much our estimate varies due to random fluctuations). The Epanechnikov kernel is chosen for its optimality in minimizing the mean integrated squared error (MISE), making it ideal for kernel density estimation. Its compact support improves computational efficiency by reducing the number of calculations, while its parabolic shape strikes a good balance between bias and variance, ensuring accurate



results (Fan and Marron 1994). It is also the default kernel in many econometric software packages, including EViews 14 (S&P Global 2024, User's Guide, p. 841).

It is important to note that the estimation of optimal bandwidth requires functional coefficient estimates for obtaining standard errors, covariance matrices, and bias estimates. These estimates depend on a pilot bandwidth, h_{pilot}, which is determined independently using several available methods (Zhang and Lee 2000) such as the simple rule of thumb, robust rule of thumb, residual squares criterion, or modified multi-cross-validation (MMCV). In our study, we use the MMCV method (Cai et al. 2000), which employs a cross-validation procedure to derive functional objective values at each evaluation point or z. The optimal pilot bandwidth h_{pilot} minimizes the sum of these objective values across all z. In this approach, parameter proliferation may occur because the objective function includes twice as many coefficients as the base model, due to the incorporation of first derivatives in the Taylor approximation. However, our primary interest usually lies in α and β rather than their respective first derivatives $\alpha^{(1)}$ and $\beta^{(1)}$.

Data and empirical findings

The Consumer Price Index was obtained from the Federal Reserve Economic Data (FRED) at https://fred.stlouisfed.org/, and average monthly ETF prices were sourced from *investing.com*. We analyzed nine sectoral ETFs along with the S&P500 index represented by SPY (see Table 1), all issued by State Street Global Advisors to ensure uniformity in definitions and to prevent constituent overlap across sectors. ETF price data begin in December 1998, so the sample period spans January 1999 to February 2024, including all available ETF price returns in the US equity market. Table 1 presents key summary statistics during the sample period, including average monthly percentage changes, standard deviation (%), skewness, kurtosis, and Jarque–Bera statistics for all ETFs: SPY (S&P 500), XLY (Consumer Discretionary), XLK (Technology), XLV (Health Care), XLF (Financials), XLU (Utilities), XLP (Consumer Staples), XLE (Energy), and others. Across various sectors, discernible trends emerged in the performance of these ETFs. For example, XLY (Consumer Discretionary), XLK (Technology), and XLV (Health Care) distinguished themselves with the highest average returns, underscoring their superior performance relative to other sector ETFs during the sample period. Conversely, XLF (Financials), XLU (Utilities), and XLP (Consumer Staples) exhibited the lowest returns during this period.

A quick glance at standard deviations in Table 1 reveals that defensive ETFs such as XLP (Consumer Staples), XLV (Health Care), and XLU (Utilities) displayed the least variability, whereas XLE (Energy), XLF (Financials), and XLK (Technology) exhibited higher levels of volatility. Of particular interest is the observation that despite XLF's (Financials) lower average monthly returns, it demonstrated the second largest standard deviation.

Table 1 Descriptive statistics (January 1999–February 2024)

Description	Mean (1)	Std. Dev. (2)	Return per unit of risk (3)=(1)÷(2)	Skewness (4)	Kurtosis (5)	Jarque-Bera (6)
Annual inflation (%)	2.340	1.645	–	1.02	5.57	135***
Monthly inflation (%)	0.212	0.306	–	−0.85	10.22	692***
Monthly ETF returns (%)						
S&P500: SPY	0.464	3.819	0.122	−1.83	11.03	979***
Materials: XLB	0.453	4.948	0.091	−1.60	10.97	928***
Energy: XLE	0.424	6.186	0.069	−2.01	15.94	2312***
Financial: XLF	0.238	5.526	0.043	−1.22	10.00	692***
Industrial: XLI	0.525	4.715	0.111	−1.62	11.70	1084***
Technology: XLK	0.606	5.331	0.114	−0.99	5.79	147***
Consumer staples: XLP	0.331	2.924	0.113	−0.95	6.16	171***
XLU: Utilities	0.233	3.912	0.060	−1.41	7.62	369***
XLV: Health care	0.572	3.473	0.165	−1.31	6.80	267***
XLY: Consumer discretionary	0.639	4.882	0.131	−1.40	9.04	558***

The table highlights that XLY, XLK, and XLV performed strongly with the highest average returns among the ETFs, while XLF, XLU, and XLP showed lower returns. Defensive ETFs such as XLP, XLV, and XLU demonstrated lower volatility, contrasting with the higher volatility of XLE, XLF, and XLK. Despite its lower average returns, XLF exhibited a notably high standard deviation. All ETFs displayed negatively skewed distributions with kurtosis values exceeding 3. Average monthly inflation (0.212%) was lower than the returns of all ETFs. SPY recorded an average monthly return of 0.464% during the study period, nearly double the inflation rate

***Indicates that the normality assumption is rejected at the 1% level

Furthermore, a consistent feature among these ETFs was their negatively skewed distribution patterns, characterized by kurtosis values exceeding 3. It is interesting to note that the average monthly inflation rate (0.212%) was lower than the monthly returns observed for all ETFs. According to Table 1, SPY (S&P 500) recorded an average monthly return of 0.464% during the study period, nearly double the monthly inflation rate. Annual inflation in the USA averaged 2.34% over the same timeframe.

Table 2 represents key characteristics of eleven prominent ETFs issued by SPDR, each tracking a specific sector of the S&P 500 Index. These ETFs are widely utilized by investors to gain targeted exposure to individual industries within the US equity market. The table provides information on each ETF's ticker, name, number of holdings, top ten largest holdings with their respective percentage weightings, and dividend yield. All the ETFs have a uniform expense ratio of 0.09%, making them cost-effective investment tools.

The number of holdings varies across the ETFs, reflecting the diverse composition of each sector. For example, XLB (Materials) consists of 28 companies, while XLF (Financials) includes 67 holdings. The top ten holdings highlight the leading companies in each sector, often accounting for a significant portion of the total assets. For instance, SPY's largest constituents include Apple and Microsoft, while XLE is predominantly weighted toward energy giants such as Exxon Mobil and Chevron. According to Table 2, dividend yields differ between sectors, with XLE (Energy) offering the highest yield at 3.7%, followed closely by XLU (Utilities) at 3.2%. XLP also offers a competitive dividend yield of 2.5%, reflecting the stability of its underlying consumer staples companies. These features make certain sector ETFs

Table 2 Key characteristics of the SPDR ETFs. *Source*: The data presented in the table are sourced from official ETF fact sheets provided by State Street Global Advisors (SSGA), along with financial data platforms such as Morningstar and Yahoo Finance, as of December 2024

Ticker	ETF name	Number of holdings	Top 10 holdings (%)	Dividend yield (%)
SPY	S&P 500	500	Apple (7.1), Microsoft (6.5), Amazon (3.1), NVIDIA (2.8), Alphabet A (1.9), Alphabet C (1.8), Tesla (1.7), Meta Platforms (1.6), Berkshire Hathaway (1.5), UnitedHealth (1.4)	1.50
XLB	Materials	28	Linde (16.8), Sherwin-Williams (6.2), Freeport-McMoRan (5.9), Air Products & Chemicals (5.6), Newmont (5.5), Ecolab (5.0), DuPont (4.6), Martin Marietta (4.4), Corteva (4.3), Vulcan Materials (4.0)	1.90
XLE	Energy	23	Exxon Mobil (23.2), Chevron (19.8), ConocoPhillips (9.1), EOG Resources (5.7), Schlumberger (4.7), Pioneer Natural Resources (4.5), Marathon Petroleum (3.9), Valero Energy (3.6), Phillips 66 (3.4), Halliburton (2.9)	3.70
XLF	Financials	67	Berkshire Hathaway (13.4), JPMorgan Chase (10.7), Visa (6.6), Mastercard (6.3), Bank of America (3.8), Wells Fargo (3.2), S&P Global (2.9), Goldman Sachs (2.7), Morgan Stanley (2.4), BlackRock (2.1)	2.00
XLI	Industrials	72	Raytheon (5.2), Honeywell (4.8), Union Pacific (4.6), Boeing (4.2), General Electric (3.9), Caterpillar (3.8), UPS (3.7), Lockheed Martin (3.4), Deere (3.3), Northrop Grumman (3.0)	1.70
XLK	Technology	75	Apple (22.4), Microsoft (21.8), NVIDIA (4.7), Broadcom (4.1), Adobe (2.7), Salesforce (2.6), Oracle (2.4), Accenture (2.1), Texas Instruments (2.0), Cisco Systems (1.8)	0.90
XLP	Consumer Staples	40	Procter & Gamble (15.6), Coca-Cola (9.9), PepsiCo (9.8), Costco Wholesale (8.3), Walmart (6.8), Mondelez (4.1), Philip Morris (4.0), Colgate-Palmolive (3.7), Estee Lauder (2.8), Kimberly-Clark (2.6)	2.50
XLU	Utilities	28	NextEra Energy (16.0), Duke Energy (8.5), Southern Co (7.4), Dominion Energy (5.9), American Electric Power (5.5), Exelon (5.3), Sempra (4.8), Consolidated Edison (4.7), Xcel Energy (4.5), Public Service Enterprise Group (4.0)	3.20
XLV	Health Care	64	UnitedHealth (9.5), Johnson & Johnson (9.1), Eli Lilly (7.1), Merck (5.8), Pfizer (4.4), Thermo Fisher (4.3), AbbVie (4.0), Danaher (3.9), Amgen (3.7), Gilead Sciences (3.2)	1.40
XLY	Consumer Discretionary	60	Amazon (24.3), Tesla (14.5), Home Depot (8.2), McDonald's (6.2), Nike (3.7), Starbucks (3.2), Booking Holdings (3.0), Lowe's (2.9), Target (2.4), TJX Companies (2.3)	0.80

This table presents key characteristics of eleven SPDR ETFs, each tracking a specific sector of the S&P 500 Index. These ETFs offer a cost-effective way for investors to gain targeted exposure to various sectors, with a uniform expense ratio of 0.09%. While the number of holdings varies across sectors, the top holdings often dominate each ETF's composition. Dividend yields differ between sectors, with energy and utility ETFs offering the highest yields



particularly appealing to income-focused investors. Overall, the table illustrates the range of investment opportunities available, whether one seeks broad market exposure through SPY or more concentrated investments in specific sectors such as XLK (Technology), XLV (Health Care), or defensive sectors like XLP (Consumer Staples) and XLU (Utilities). These ETFs enable investors to tailor their portfolios according to specific goals, balancing growth potential, income generation, and risk exposure.

Table 3 presents the optimal bandwidth estimates are obtained by minimizing the objective function in Eq. (4), highlighted in bold, using the multi-cross-validation method. Determining the optimal bandwidth using the multi-cross-validation method is a crucial step in nonparametric data analysis, particularly in kernel-based smoothing techniques. In this study, the optimal bandwidth is derived by minimizing an objective function, which helps identify the best smoothing parameter that balances the bias–variance trade-off. The multi-cross-validation method is used to systematically test different bandwidth values and choose the one that minimizes prediction errors. By doing so, it ensures that the model's smoothing process is neither too rough (high bias) nor too sensitive to noise in the data (high variance). The usefulness of this approach lies in its ability to fine-tune the smoothing process, enabling the model to capture underlying patterns without overfitting or underfitting the data. The multi-cross-validation method is particularly valuable because it evaluates the model's performance across different subsets of the data, providing a more robust estimate of the optimal bandwidth. In Table 3, the optimal bandwidth estimates are calculated using a second-degree polynomial combined with the Epanechnikov kernel, which is commonly used in the literature due to its efficient bias–variance trade-off. This approach enhances the reliability and generalizability of the results.

Figure 1 presents the estimated functional coefficients graphically over the sample period from January 1999 to February 2024, using 302 monthly observations. The individual scatter plots depict the relationship between the estimated β (y-axis) and annualized inflation rates (x-axis) for each of the nine sectoral ETFs. The y-axis also shows the distribution of the estimated beta coefficients. The

Table 3 Estimated final bandwidth using the multi-cross-validation method

Bandwidth	XLB	XLE	XLF	XLI	XLK	XLP	XLU	XLV	XLY
0.1000	0.00114	0.00403	0.00061	0.00082	0.00061	0.00279	0.00127	0.00070	0.00045
0.1100	0.00094	0.00361	0.00056	0.00068	0.00051	0.00224	0.00127	0.00072	0.00040
0.1210	0.00087	0.00338	**0.00051**	0.00066	0.00050	0.00204	0.00129	0.00075	0.00036
0.1331	0.00035	0.00156	0.00052	0.00035	0.00045	0.00075	**0.00082**	0.00039	0.00029
0.1464	0.00032	0.00159	0.00056	0.00031	0.00052	0.00056	0.00100	0.00039	0.00026
0.1611	0.00032	0.00156	0.00058	0.00031	0.00045	0.00052	0.00123	0.00042	0.00025
0.1772	0.00031	0.00151	0.00058	0.00031	0.00043	0.00051	0.00133	0.00042	0.00023
0.1949	0.00030	0.00147	0.00056	0.00030	0.00041	0.00052	0.00129	0.00041	0.00022
0.2144	0.00030	0.00143	0.00056	0.00030	0.00038	0.00051	0.00118	0.00040	0.00021
0.2358	0.00029	0.00145	0.00056	0.00029	0.00036	0.00050	0.00118	0.00039	0.00020
0.2594	0.00029	0.00148	0.00057	0.00028	0.00036	0.00050	0.00117	0.00039	0.00019
0.2853	0.00029	0.00147	0.00058	0.00028	0.00035	0.00050	0.00115	0.00039	0.00019
0.3138	0.00029	0.00138	0.00056	0.00027	0.00034	0.00050	0.00114	0.00038	0.00018
0.3452	0.00029	0.00138	0.00056	0.00027	0.00034	0.00051	0.00114	0.00038	0.00018
0.3798	0.00030	0.00139	0.00056	0.00026	0.00035	0.00051	0.00112	0.00038	0.00017
0.4177	0.00030	0.00138	0.00055	0.00026	0.00034	0.00051	0.00108	0.00038	0.00017
0.4595	0.00030	0.00139	0.00055	0.00025	0.00033	0.00051	0.00106	0.00038	0.00017
0.5054	0.00029	0.00138	0.00054	0.00025	0.00033	0.00051	0.00104	0.00037	0.00017
0.5560	0.00029	0.00137	0.00054	0.00025	0.00032	0.00051	0.00104	0.00037	**0.00017**
0.6116	0.00028	0.00137	0.00054	0.00024	0.00032	0.00050	0.00103	0.00036	0.00017
0.6728	0.00027	0.00136	0.00053	0.00024	0.00032	0.00050	0.00103	0.00036	0.00017
0.7400	**0.00026**	**0.00133**	0.00054	0.00024	0.00032	0.00050	0.00102	**0.00035**	0.00018
0.8140	0.00151	0.01286	0.02353	0.00057	0.00042	0.00068	0.00441	0.00268	0.00419
0.8954	0.00044	0.00256	0.00354	**0.00022**	**0.00029**	**0.00038**	0.00127	0.00067	0.00045
0.9850	0.00046	0.00337	0.00162	0.00037	0.00033	0.00040	0.00128	0.00052	0.00038

This table presents the final bandwidth estimates determined by minimizing the objective function, highlighted in bold, using the multi-cross-validation method. This estimation employs a second-degree polynomial with the Epanechnikov kernel, which is the most widely used kernel in the literature



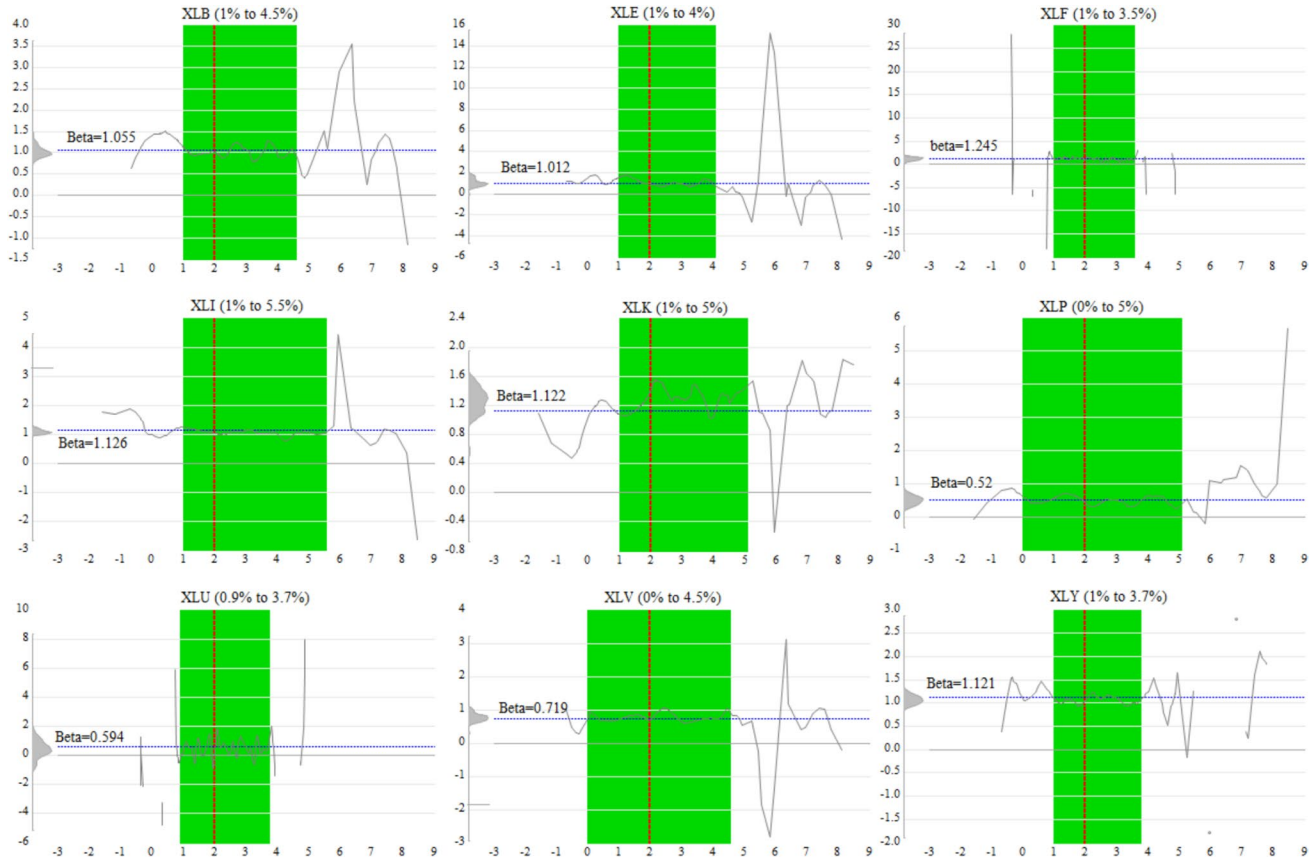


Fig. 1 Relationship between estimated beta and annualized inflation rates by sector (January 1999–February 2024) Note: The individual scatter plots depict the relationship between estimated beta (*y*-axis) and annualized inflation rates (*x*-axis) for each of the nine sectoral ETFs during the period January 1999 to February 2024. The highlighted green area indicates when beta remains a stable and reliable measure for financial analysts within the specified range. It can be observed that this range is widest for XLP (0–5%, as a defensive ETF) and narrowest for XLY (1–3.7%, as the most inflation-sensitive ETF). Once inflation exceeds the highlighted range, betas exhibit erratic behavior across almost all sectors, rendering them an unreliable measure of systemic risk. This highlights central bankers' strict commitment to maintaining inflation around 2% over business cycles, as indicated by the vertical red dotted lines.

highlighted green area indicates when β remains a stable and reliable measure within the specified range. As can be seen, this range is widest for XLP (0–5%, as a defensive ETF) and narrowest for XLY (1–3.7%, as the most inflation-sensitive ETF). The horizontal blue dotted lines represent the point estimate for β, typically passing roughly through the middle of the gray distributions on the *y*-axis, indicated by gray bell shapes. The lowest estimated β belongs to the three defensive ETFs (i.e., XLP = 0.52, XLU = 0.594, XLV = 0.719), while the highest β is seen in XLF = 1.245, XLI = 1.126, and XLK = 1.122 (see the horizontal blue dotted lines).

However, as soon as annualized inflation rates exceed the corresponding highlighted green range, β exhibits erratic and explosive behavior across all sectors (see the gray solid lines), rendering them an unreliable measure of systemic risk for investment strategists. These results underscore the importance of maintaining inflation around 2% in the USA, indicated by the vertical red dotted line in each graph. *Inter alia*, this explains why the Fed is so vigilant in its fight against inflation. High inflation erodes consumers' purchasing power, reducing demand for goods and services, which lowers company profits and stock prices. It also leads to higher interest rates, increasing borrowing costs and hindering business investment and expansion, further slowing economic growth and negatively impacting stock valuations. All of this undermines the concept of systematic and sector-specific risk, which are key elements of sectoral portfolio rebalancing over time. A quick glance at the individual graphs in Fig. 1 reveals that, as a rule of thumb, β begins deviating outside the highlighted green area, typically when annual inflation exceeds the 1–5% band.

While Fig. 1 illustrates the relationship between estimated beta and annualized inflation rates for each sector, one may also be interested in how betas evolve over time. To this end, Fig. 2 presents the recursive beta estimates by sector for the post-GFC period (2010m1–2024m4), offering a dynamic view of how betas change over time. The estimation process begins by calculating the first beta using the minimum

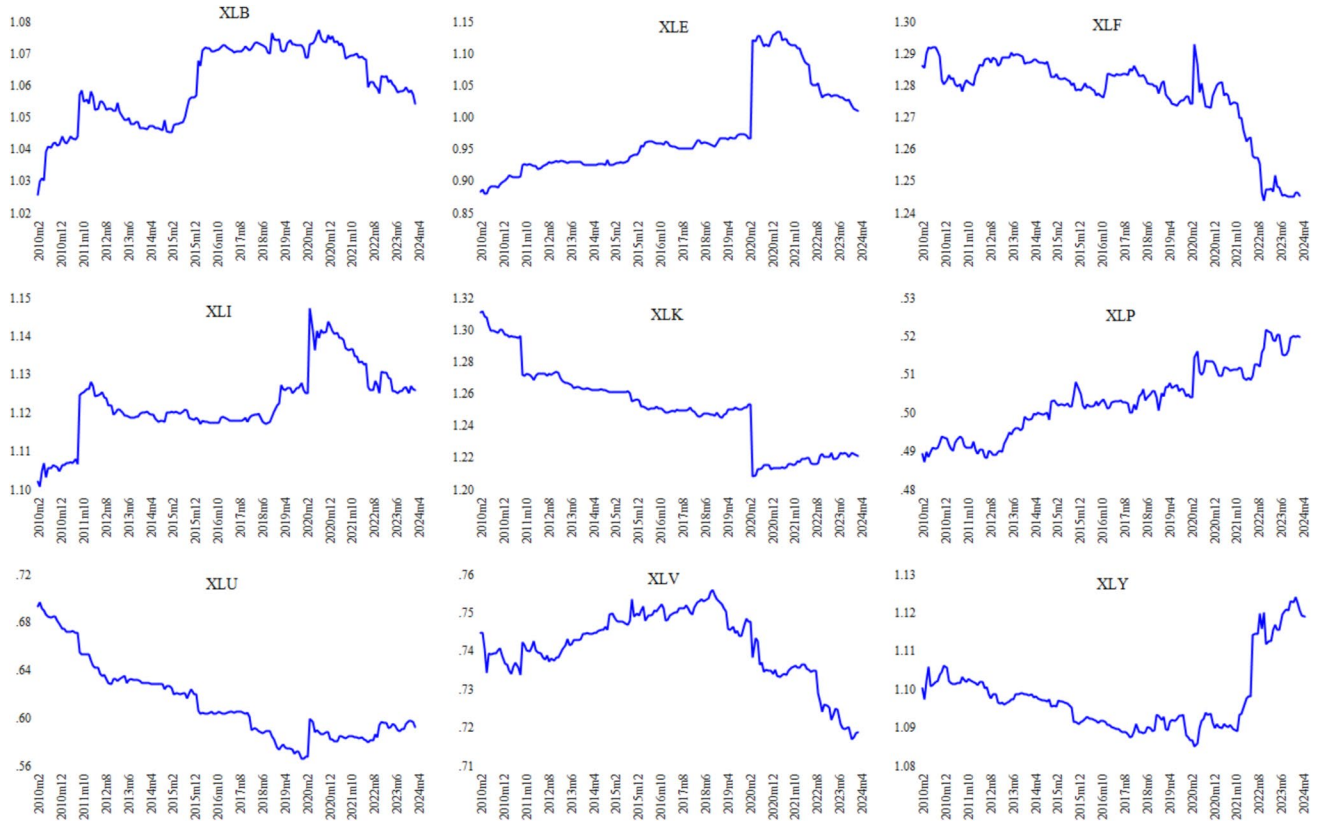


Fig. 2 Time-varying recursive beta estimates by sector (January 2010–April 2024) Note: These graphs present the recursive beta estimates for each sector during the post-GFC period (2010m1–2024m4), showing how sector betas evolve over time. For example, the betas for the Technology (XLK) and Financials (XLF) sectors exhibit a declining trend, indicating reduced sensitivity to market movements. In contrast, the beta for the Consumer Staples sector (XLP) is rising, suggesting increased responsiveness to market changes.

required sample period, where monthly returns for a given sector are regressed against an intercept and the monthly returns of the market (SPY). The recursive estimation process then proceeds iteratively: at each step, one additional month of data is added to the sample until the final beta is computed using the full sample. The concept of beta here refers to the sensitivity of a sector's returns to changes in the overall market returns, reflecting the sector's systematic risk. A higher beta indicates that the sector is more sensitive to market movements, while a lower beta suggests less sensitivity. The recursive estimation approach allows for the examination of how beta evolves over time, providing insights into the changing relationship between sector returns and market movements.

As shown in the individual graphs in Fig. 2, sector betas tend to fluctuate over time. For example, the recursive betas for the Technology sector (XLK), Utilities sector (XLU), and Financials sector (XLF) exhibit an overall declining trend, suggesting these sectors have become less sensitive to market movements in recent years. During the post-COVID-19 pandemic, when inflation was relatively high, the individual graphs in Fig. 2 reveal that the betas started falling for all sector ETFs except XLP (Consumer Staples) and XLY (Consumer Discretionary), which began rising. XLK (Technology) started rising at a much slower pace. In the post-COVID-19 period, the time-varying beta for the Consumer Discretionary sector (XLY) has risen, reflecting increased market sensitivity, while betas for sectors such as Healthcare (XLV), Industrials (XLI), Energy (XLE), and Materials (XLB) have fallen, indicating a decrease in their sensitivity to market fluctuations.

Conclusion

This paper specifies and estimates the monthly β coefficient as a function of annualized inflation for all SPDR sector ETFs using data from January 1999 to February 2024. As expected, the defensive ETFs (XLP, XLU, and XLV) exhibit lower betas and generally have a higher tolerance for inflation, indicated by the wider green shade in Fig. 1. In contrast, ETFs such as XLY, XLF, and XLE are more sensitive to inflation fluctuations. However, once inflation moves out of the 1–5% range, particularly beyond 4–5%, the functional

β for all nine sectoral ETFs display unstable and nonsensical behavior. This deviation signifies that none of the sectoral ETFs are immune to the adverse effects of high inflation, underscoring the importance of keeping inflation within a manageable range to ensure the reliability of beta as a measure of systemic risk. Unstable systemic risk presents a challenging environment for investment strategists, akin to targeting a moving object, where higher inflation unpredictably impacts sectoral returns relative to the market.

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Declarations

Conflict of interest The author is not aware of any conflict of interest that may affect the impartiality of his analysis.

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